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To the Editor of *Agricultural Systems*

Dear Editor,

I submit the manuscript **"Light-sharing control for tracker-based agrivoltaics under a yield floor: what learning adds and what shade-response uncertainty costs"** for consideration as an original research article in *Agricultural Systems*.

What the paper does. Agrivoltaic systems put crops and photovoltaic modules on the same land. Regulators and farmers increasingly require the crop to keep most of its reference yield, so the practical question is how much electricity a farm can produce under such a yield floor. The paper treats this as a systems problem: five staple crops (wheat, rice, maize, soybean, potato) on 400 crop–site pairs sampled in proportion to their harvested area, four climates, and a differentiable simulator that couples tracker geometry, PV output, light and microclimate under the modules, and crop growth. The array density of every site is chosen from its own history only, and performance is measured on held-out sites and years.

Fit to the journal. The study analyses an interaction between food and energy systems on farmland, at farm and landscape level (the density and control of the array, the crop, the climate, and the yield-retention floor). It is a decision problem under uncertainty, where the uncertainty is the crop response to shade, and it uses simulation, a learned controller and learned design rules, which falls within the journal's scope on systems modelling, decision-making and artificial intelligence in agricultural systems.

What it adds.

- It separates how much of the electricity under a yield floor comes from array density and how much from daily light-sharing control: light-sharing rules allow 2.1 to 2.9 times the module density of static single-axis arrays and raise the land-equivalent ratio from 1.07 to 1.18–1.24, but the floor is missed in 24–31% of seasons.
- It tests what learning adds, and reports that honestly. A neural controller trained through the simulator exceeds a written phenology rule by 0.044 LER, but an optimised window by only 0.007, and the difference is not distinguishable from zero at matched retention.
- It treats the crop response to shade as a distribution. A model fitted to 428 observations from 58 studies predicts a new study no better than proportional loss for the staple crop types, and the simulator is milder than the data for maize. A density designed under the mean response leaves 70% of pairs at the floor when another plausible response holds, against 87% for an oracle design. Conformal margins attain a stated compliance inside the ensemble of responses, a learned design network does worse than a simple margin, and only large margins are safe outside the ensemble.
- It reports the economics and the scale as sensitivity analyses: under illustrative costs light sharing breaks even with the open field at 70–92 USD per MWh, static arrays at 59 and a conventional plant at 48, and agrivoltaics on 5% of staple cropland would supply about 14 PWh per year only with about 10 TWp of modules.

Limits stated in the paper. All results come from a simulator. There is no field validation under trackers, the shade response of maize is uncertain, the economic parameters are assumptions, and the 90% floor is a modelling convention. The main text is about 5,700 words; the full methods and further analyses are in the supplementary material.

Declarations. The work is original, has not been published, and is not under consideration elsewhere. The declaration of competing interests is provided in the submission system. All data are public; the code, the site list and the result tables are available in a public repository, and a DOI-registered archive will be cited before publication. The submission includes the manuscript, highlights, graphical abstract and supplementary material.

Thank you for considering the manuscript.

Sincerely,

Quang-Vinh Dang British University Vietnam